



Day of Data Toronto 2026



Toronto Data Professionals Community (TDPC)



Day of Data Toronto

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[Toronto Data Professionals Community \(TDPC\)](#), one of the largest data professional's community in Toronto, host monthly event which offers interactive learning built by community and guided by trusted data experts.

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Toronto Data Professionals Community (TDPC)



SQL Saturday (#1160)

Why AI Projects Fail

Data Platform Lessons Every Architect Should Know

Day of Data Toronto · 26 September 2026

Daniele Mario Areddu



Toronto Data Professionals Community (TDPC)



SQL Saturday (#1160)

The perspective behind this talk

Daniele Mario Areddu

Backend & AI Developer · Global Technologies Italia

**Building AI systems whose
answers can be defended.**

Computer Science student · University of Calabria

International technical speaker on AI, backend systems,
and data platforms



Toronto Data Professionals Community (TDPC)



SQL Saturday (#1160)

Natural language removes a real barrier to data access

HYPOTHETICAL DISTRIBUTOR

**One interface.
Three different decisions.**

Revenue trend / Order status / Inventory
promise



Illustrative facility · unrelated to the hypothetical distributor

The platform already makes diverse data reachable

01

SQL Server / Azure SQL

Transactional records and operational state

02

Fabric analytical estate

Prepared histories and cross-domain analysis

03

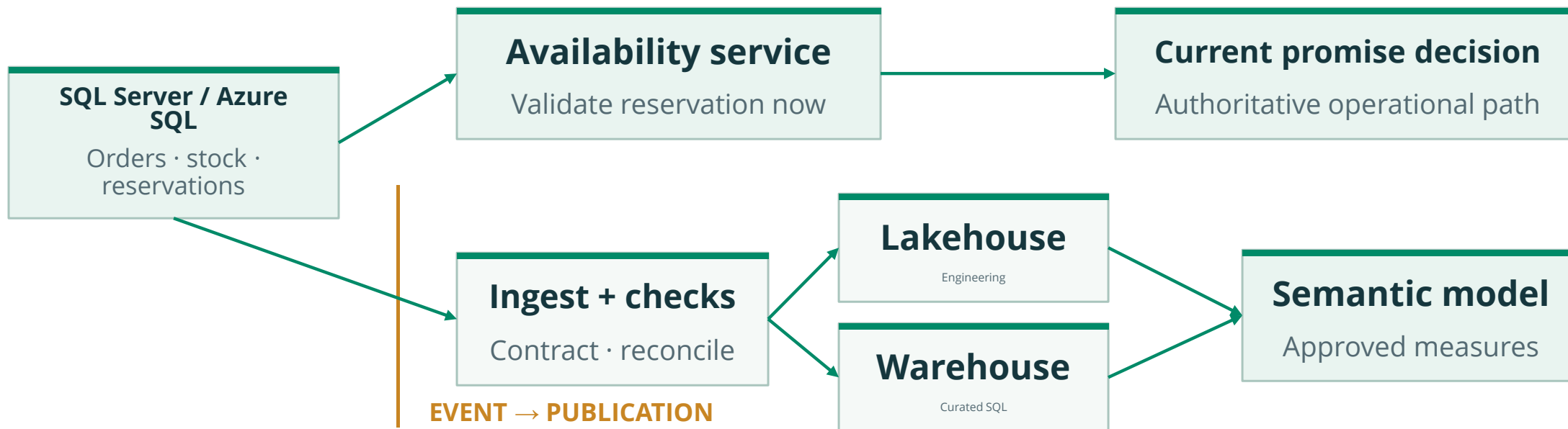
Semantic model

Reusable measures and business relationships



Operational and analytical systems answer different questions

FULFILLMENT DOMAIN · owns current availability



Fabric provides choices, not a universal data surface

LAKEHOUSE

- Flexible engineering and preparation
- Files and tables for iterative transformation

WAREHOUSE

- Curated relational analysis
- Governed SQL-oriented consumption

A semantic model is the business-facing measure layer above either path.



Semantic models can make business questions repeatable

A measure is a decision about meaning, not just arithmetic.

Revenue • recognized or booked? | Orders • placed or fulfilled?



The credible promise is access with established meaning

What would a useful answer have to contain?

A named, authorized source

The applicable business definition

A time boundary and enough evidence to check it



The interface hides distinctions the architecture must preserve

“Available inventory”

On hand

Unreserved

Sellable

One phrase can conceal three owners, clocks, and decision rules.



Each boundary can discard evidence the next layer needs



Lost on the way: event cutoff • reconciliation status • decision scope

Authority depends on the question being asked

Claim	Authoritative route	Decision window
Recent order status	Operational order system	Current workflow
Recognized revenue	Approved finance product	Published period
Inventory promise	Operational availability	At commitment



Freshness and consistency are different properties

EVENT TIME

ARRIVAL

PUBLICATION



A stock movement occurs



Related records may lag



A reconciled snapshot is released



A quality failure travels into the final sentence

ILLUSTRATIVE PRODUCT CONTRACT · Fulfillment / v2

SCHEMA	Source event ID and product key must remain mapped
---------------	--

QUALITY	Unique events; valid product keys; reconciled totals
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ELIGIBILITY	Failed publication → affected answers ineligible
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Owner approves changed meaning; platform gates publication.



Common storage does not create common meaning

Same table location $\neq$ same business definition.

Meaning needs an owner, scope, version, and approved consumers.



Two revenue measures can both be correct

SALES VIEW

- Booked order value
- Useful for commercial demand

FINANCE VIEW

- Recognized revenue
- Useful for financial reporting

The assistant must name the measure and its owner before comparing periods.



Context engineering decides what counts as evidence

01

**Which evidence
route?**

02

**Under whose
identity?**

03

**How recent and
traceable?**



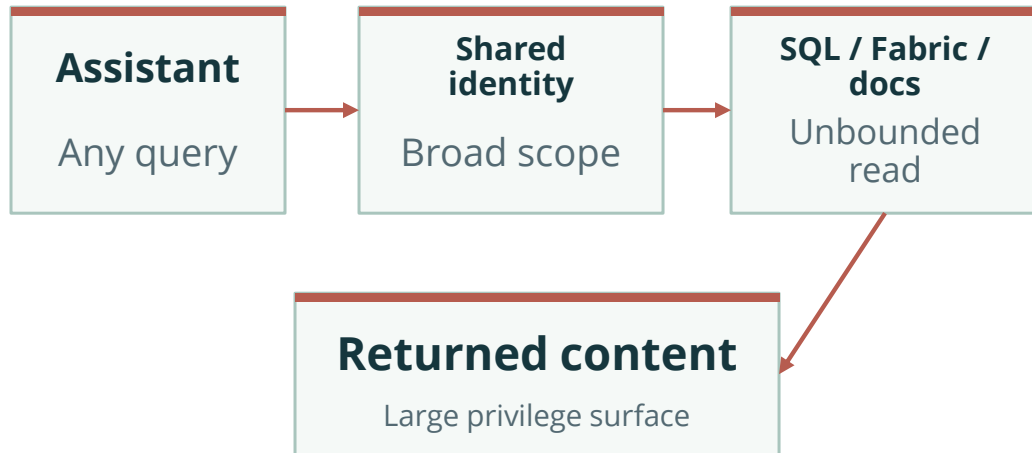
Structured facts and documents need different routes

Question type	Evidence route	Necessary control
Current numeric state	Structured, approved query	Time + identity
Policy interpretation	Versioned document retrieval	Citation + scope
Combined explanation	Both routes, reconciled	Provenance of each

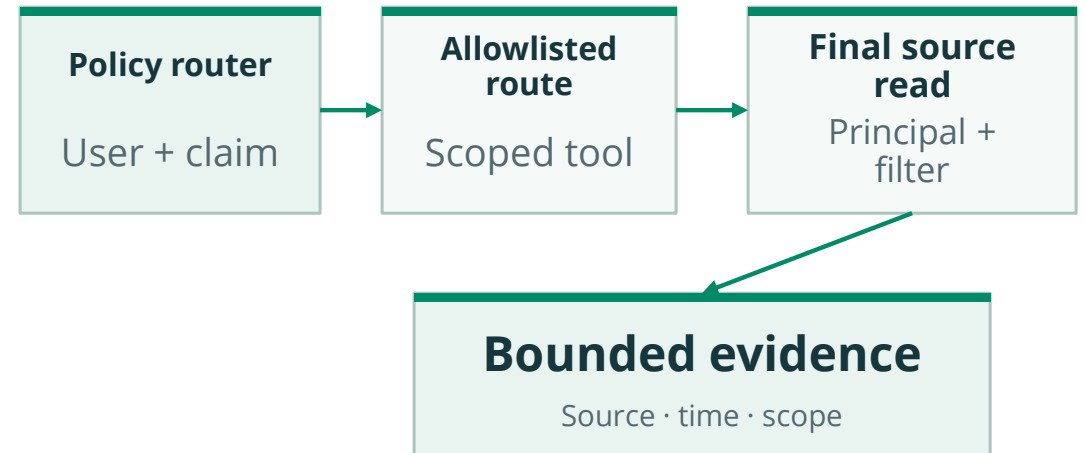


Broad assistant access buys flexibility at a governance cost

BROAD · FAST TO EXPLORE

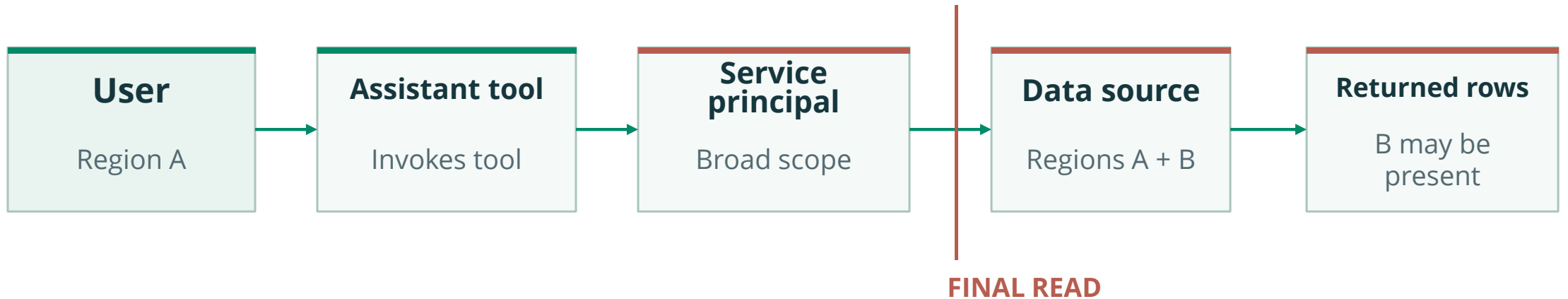


MEDIATED · ENGINEERING COST, BOUNDED READ



Permissions follow execution paths, not diagram intent

HYPOTHETICAL FAILURE PATH / REGION A USER



Chat login cannot authorize a downstream read made with broader credentials.

An answer is a governed evidence product

**Claim + definition + authorized evidence + time +
provenance**

When one element is missing, change the form of the answer.



Domain ownership and platform standards must coexist

DOMAIN OWNER

- Metric meaning; decision tolerance
- Publication approval; exception policy

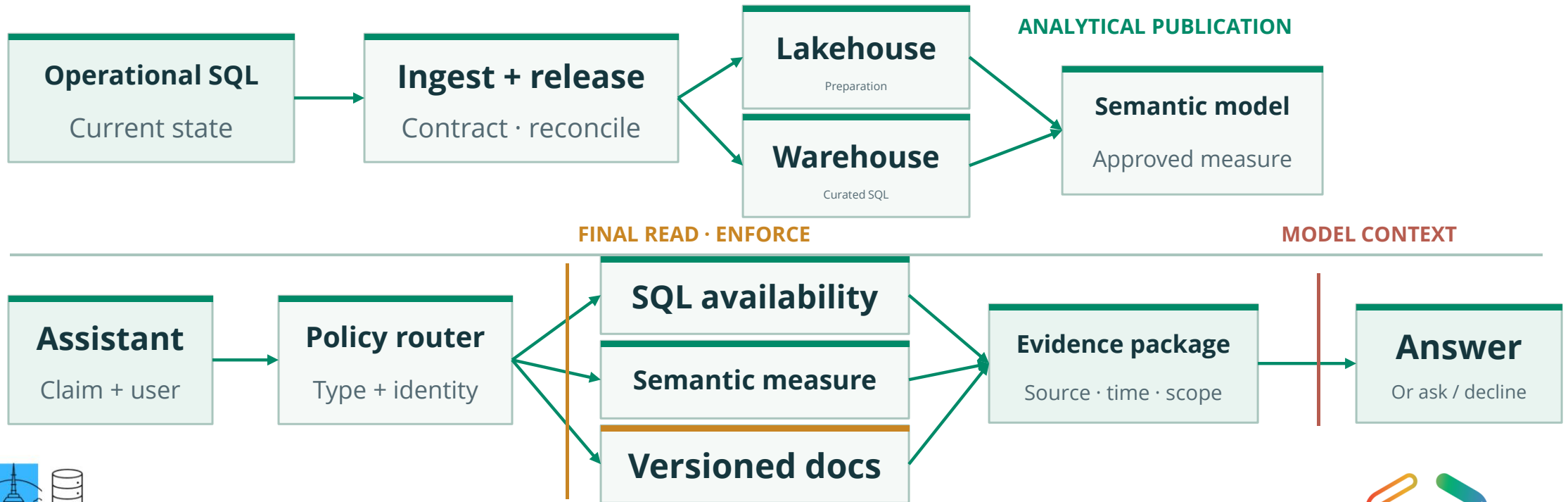
PLATFORM TEAM

- Ingestion and reconciliation
- Identity paths; observability; reliability



A governed evidence layer preserves the right boundaries

DOMAIN OWNERS APPROVE · claim, definition, scope, decision window



Freshness is a policy about the decision

How old may this answer be?

Inventory promise: ask an operational source or decline

Revenue trend: use a published, reconciled period

Show “as of” and the applicable decision window



Authorization must be tested at the final data read

The policy is route specific; verify the effective principal at each read.

OPERATIONAL SQL

Constrained query service

Approved row scope

ANALYTICAL MODEL

User / configured identity

Model permissions + RLS

DOCUMENT RETRIEVAL

Indexed principal + filters

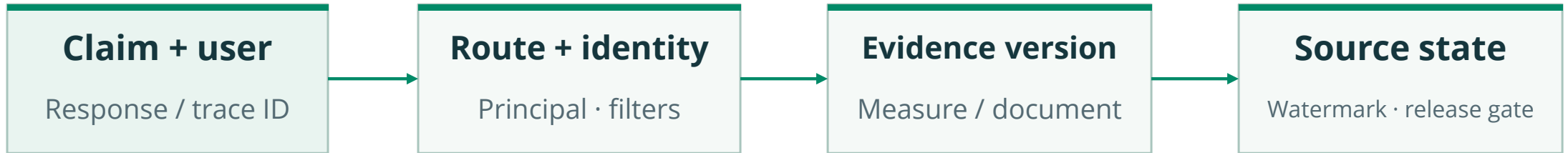
Version + document scope

Check denied reads and cache reuse across users.



Trace an answer back through its dependencies

APPLICATION ANSWER TRACE · designed and operated by the assistant team



FABRIC ITEM LINEAGE

Platform dependencies help diagnosis; the answer trace above is a separate design.



Every improvement spends a budget somewhere

Answer class	Admission gate	Budget / failure mode
Inventory decision	Current reservation check	Latency + source load
Revenue trend	Reconciled metric version	Publication lag + cost
Policy explanation	Scoped document version	Index update + exposure



Trustworthy answers need a release process across teams

DEFINE

Version meaning and data contracts

EVALUATE

Evaluate answer paths and denied cases

PUBLISH

Reconcile and certify publication

RESPOND

Assign incident ownership and rollback



A stock promise passes six gates or is withheld

“Can we promise this item now?” · HYPOTHETICAL DISTRIBUTOR

SOURCE

Operational availability + reservation

TIME / CONSISTENCY

Current, coherent state at commitment

TRACE

Source time, rule and decision recorded

OWNER / MEANING

Fulfillment defines available to promise

IDENTITY

Authorized region at final read

EXIT

Allow if verified · clarify scope · decline if unsafe



Which claim would your architecture refuse to make?

Questions & answers

LET'S CONNECT



GitHub

DaniAreddu



LinkedIn

Daniele Mario Areddu



Sessionize

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LINKEDIN · SCAN TO CONNECT



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